Smoothing ladder of the best evolved design

`re_ec3i042_r01_p10_c56`: $W_{\mathrm{ULTRALIGHT_v1}} = 80.623796$ as exported, seven smoothing levels

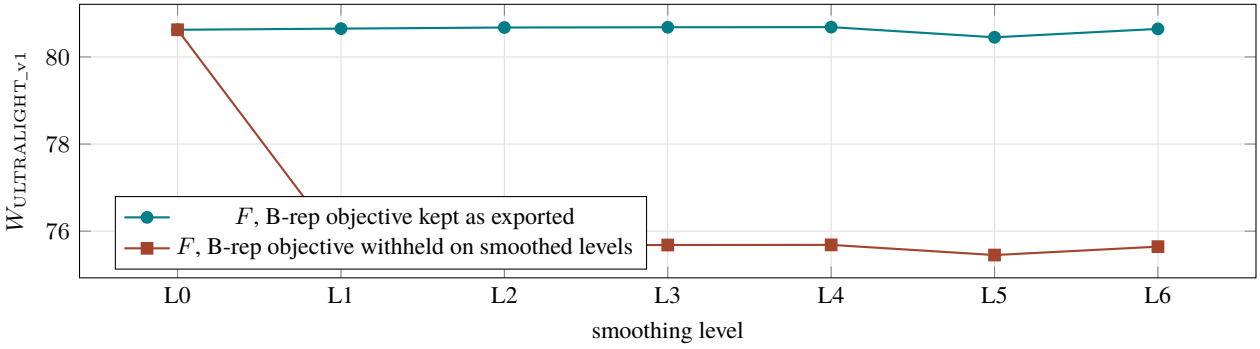
Research draft; author approval pending. Generated by `make-Cdocs/papersmoothing-ladder`; receipt `docs/paper/ladder/ladder.json`.

What this shows. The exported triangle mesh of the design is smoothed at seven strengths, from none (L0) to the strongest setting the smoother completes on its knife edges without reversing a face (L6). Each level is re-scored with the canonical `W_ULTRALIGHT_v1` and rendered from one fixed oblique camera under one low raking light, with the thickness direction exaggerated and faces coloured by height, so that grinds, serrations, channels and faceting stand out. Every camera, light, colour range and exaggeration is shared by all levels; image differences are mesh differences.

Smoothing. Taubin λ/μ filtering ($\lambda = 0.5$, $\mu = -0.5$), keeping only the component of each umbrella vector along the area-weighted vertex normal, total displacement capped at 2 mm. Vertices on edges whose two face normals differ by more than the pin angle do not move. L1–L5 pin at 30° (L4, 20 iterations, is the R1 setting of paper variant B); L6 pins at 60°. With 60° pins, 60 or more iterations reverse a face on this mesh and the smoother stops (fail-closed), so the ladder ends at L6; the previous best design, `w406t3_r01_p10_c16`, completed three more levels (60, 120 and 200 iterations).

Re-scoring. Three of the twenty objectives read the exported 3D geometry and are re-measured at every level: 3D mass (enclosed volume times $4.43\,\mathrm{g\,cm^{-3}}$), the solid tip (scikit-fem P2 submodel on the smoothed skin, cross-checked by DOLFINx 0.9.0 on the same mesh) and the CAD/B-rep objective. The B-rep objective is withheld (scored 0) on every smoothed level, because a smoothed triangle mesh has no validated STEP solid; that costs 5 points at once, so the table also gives F with the B-rep objective kept as exported, which isolates the effect of smoothing. The other seventeen objectives are computed from the genome and carry over unchanged: in particular, edge sharpness and point profile are *not* re-measured on the smoothed mesh. Neither score is a physical test.

Level	Pins	Iter.	Max disp. (mm)	RMS disp. (mm)	Vol. (%)	Mass (g)	Tip SF (DOLFINx)	F	F (B-rep kept)
L0	–	0	–	–	–	83.5536	0.6886	80.623796	80.623796
L1	30°	2	0.087	0.0058	-0.014	83.5420	0.6964	75.650199	80.650199
L2	30°	5	0.144	0.0096	-0.031	83.5277	0.7036	75.675500	80.675500
L3	30°	10	0.177	0.0127	-0.054	83.5083	0.7040	75.681721	80.681721
L4	30°	20	0.218	0.0161	-0.088	83.4803	0.7026	75.684768	80.684768
L5	30°	60	0.297	0.0231	-0.149	83.4291	0.6203	75.451120	80.451120
L6	60°	20	0.503	0.0366	0.006	83.5585	0.6957	75.643828	80.643828



Colour scales. Height above the mid-plane, 0 to the stock half-thickness 2.62 mm (thicker parts such as the finger ring saturate):



0 mm 2.62 mm

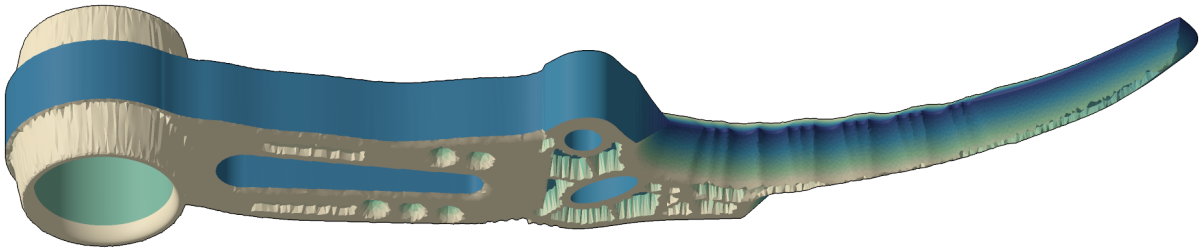
Displacement from L0, shared by all levels, 0 to 0.365 mm:



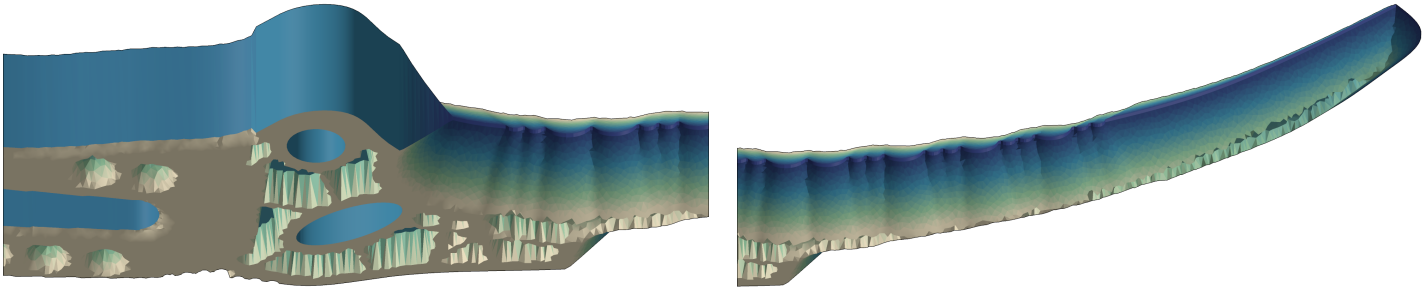
0 mm 0.365 mm

Level L0 as exported (no smoothing)

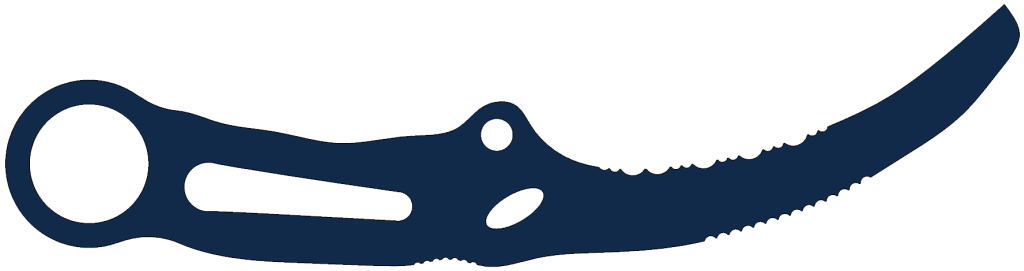
$F = 80.623796$ (with B-rep kept: 80.623796) max displacement 0.000 mm, RMS 0.0000 mm tip SF 0.6886 3D mass 83.5536 g



Whole design, oblique view, raking light, thickness direction exaggerated 2.5×; colour is height above the mid-plane (scale on the summary page).



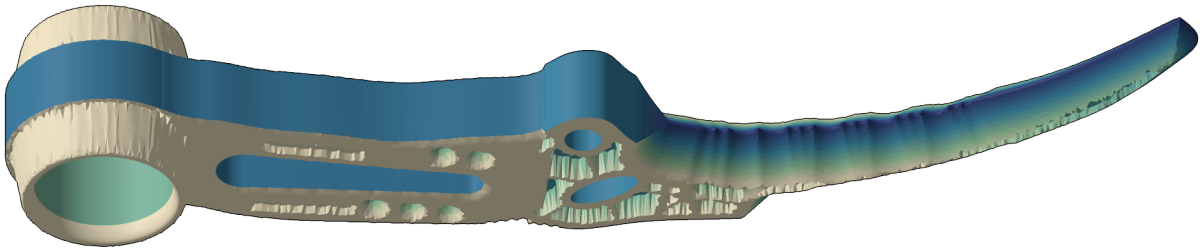
Close-ups with the same camera rotation, light and colour: guard and pivot zone (isogrid pockets, fuller windows), left; blade (grinds, serrations, tip), right.



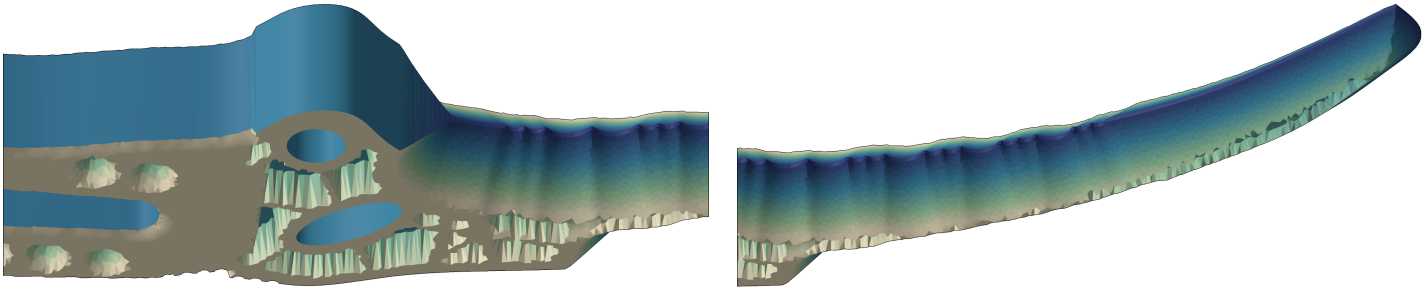
Top view; colour is displacement from L0 on the shared scale (summary page). L0 is zero everywhere.

Level L1 2 iterations, feature pins 30°

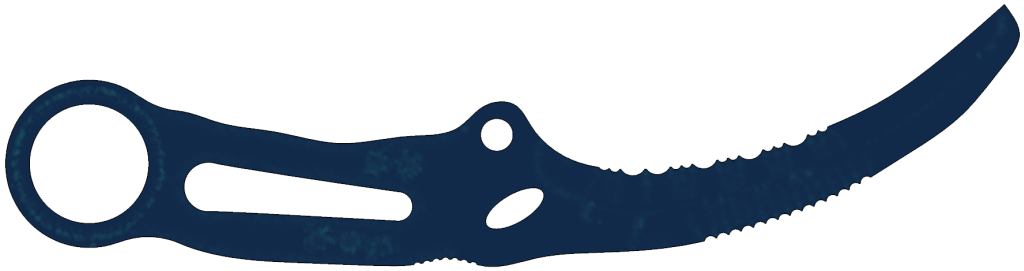
$F = 75.650199$ (with B-rep kept: 80.650199) max displacement 0.087 mm, RMS 0.0058 mm tip SF 0.6964 3D mass 83.5420 g



Whole design, oblique view, raking light, thickness direction exaggerated 2.5×; colour is height above the mid-plane (scale on the summary page).



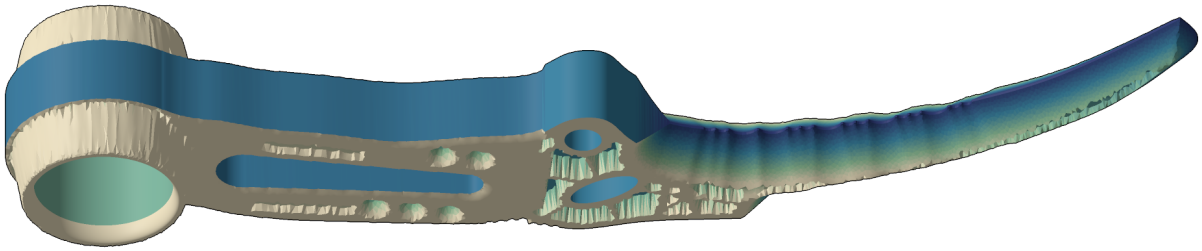
Close-ups with the same camera rotation, light and colour: guard and pivot zone (isogrid pockets, fuller windows), left; blade (grinds, serrations, tip), right.



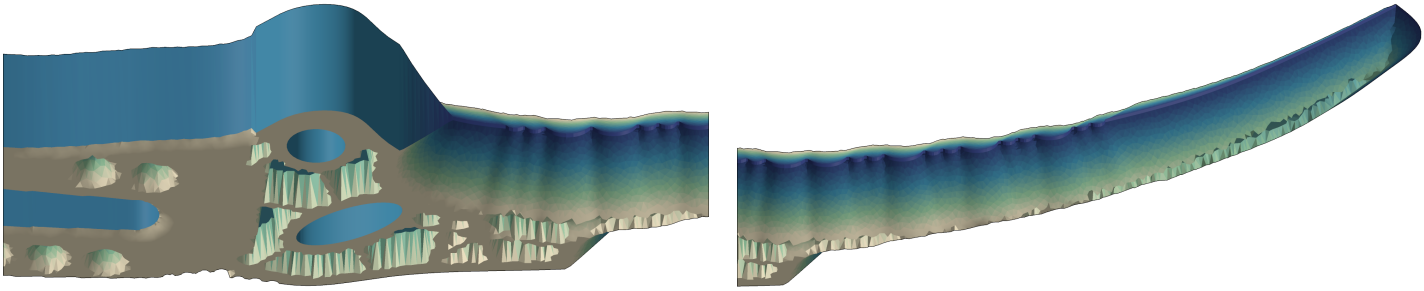
Top view; colour is displacement from L0 on the shared scale (summary page). L0 is zero everywhere.

Level L2 5 iterations, feature pins 30°

$F = 75.675500$ (with B-rep kept: 80.675500) max displacement 0.144 mm, RMS 0.0096 mm tip SF 0.7036 3D mass 83.5277 g



Whole design, oblique view, raking light, thickness direction exaggerated 2.5×; colour is height above the mid-plane (scale on the summary page).



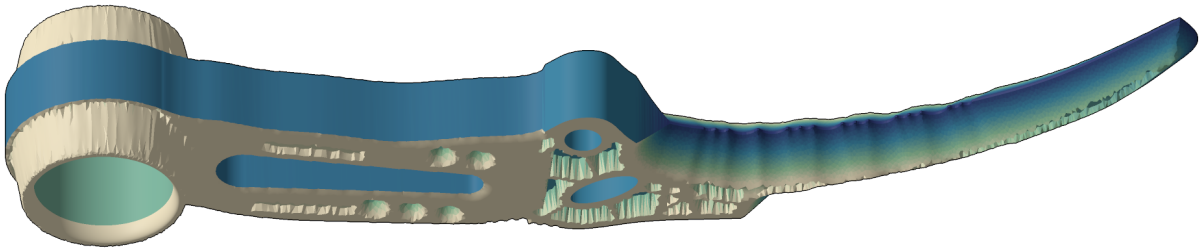
Close-ups with the same camera rotation, light and colour: guard and pivot zone (isogrid pockets, fuller windows), left; blade (grinds, serrations, tip), right.



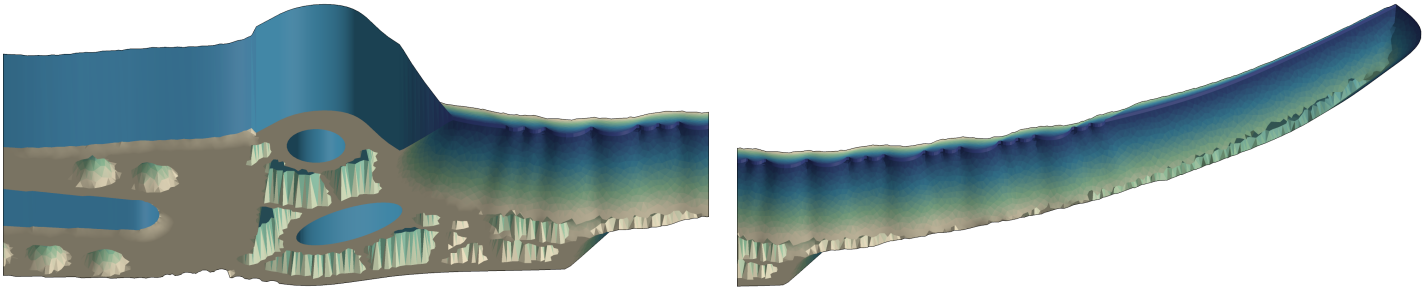
Top view; colour is displacement from L0 on the shared scale (summary page). L0 is zero everywhere.

Level L3 10 iterations, feature pins 30°

$F = 75.681721$ (with B-rep kept: 80.681721) max displacement 0.177 mm, RMS 0.0127 mm tip SF 0.7040 3D mass 83.5083 g



Whole design, oblique view, raking light, thickness direction exaggerated 2.5×; colour is height above the mid-plane (scale on the summary page).



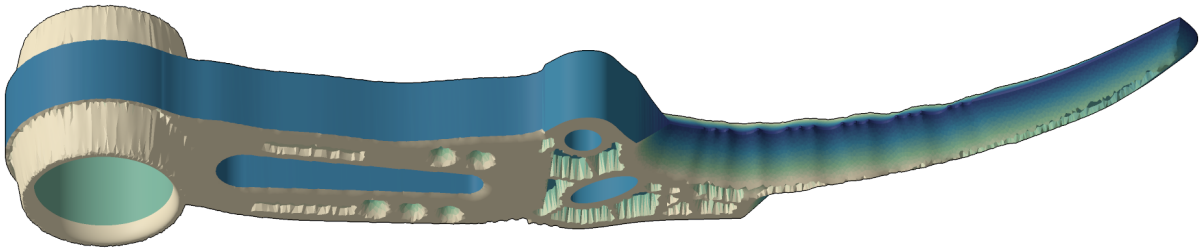
Close-ups with the same camera rotation, light and colour: guard and pivot zone (isogrid pockets, fuller windows), left; blade (grinds, serrations, tip), right.



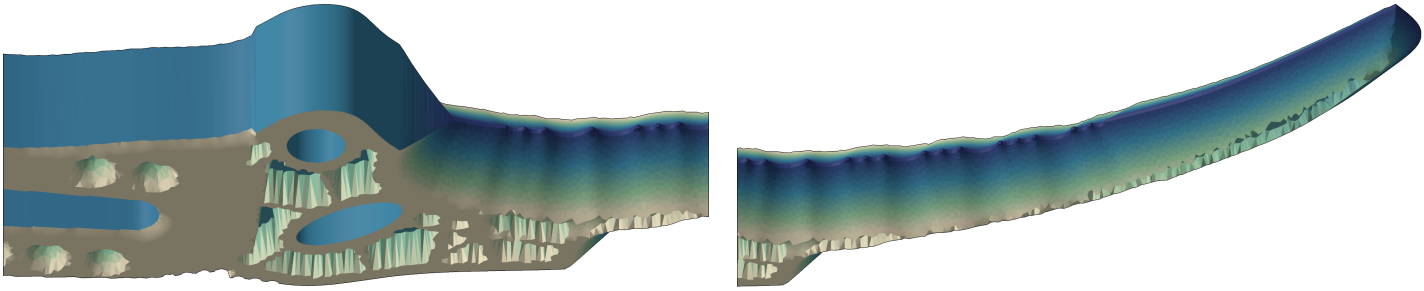
Top view; colour is displacement from L0 on the shared scale (summary page). L0 is zero everywhere.

Level L4 20 iterations, feature pins 30°

$F = 75.684768$ (with B-rep kept: 80.684768) max displacement 0.218 mm, RMS 0.0161 mm tip SF 0.7026 3D mass 83.4803 g



Whole design, oblique view, raking light, thickness direction exaggerated 2.5×; colour is height above the mid-plane (scale on the summary page).



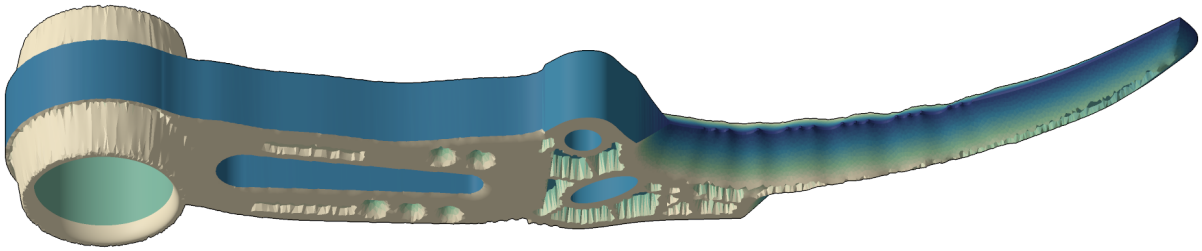
Close-ups with the same camera rotation, light and colour: guard and pivot zone (isogrid pockets, fuller windows), left; blade (grinds, serrations, tip), right.



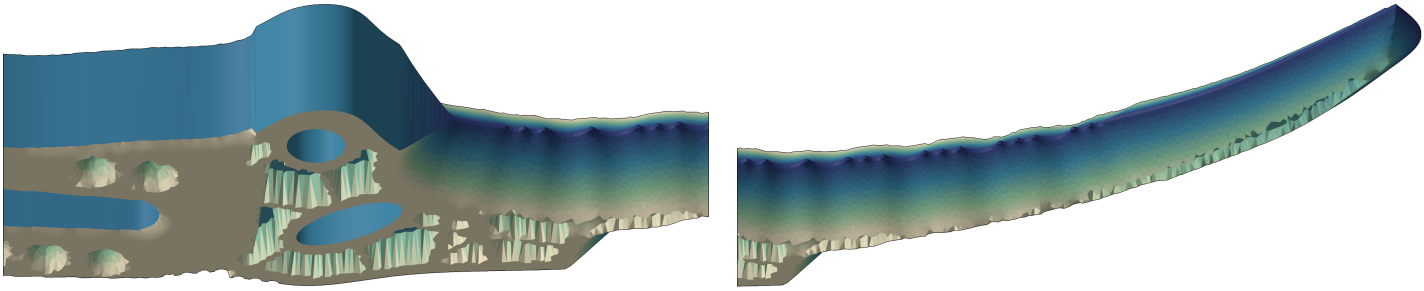
Top view; colour is displacement from L0 on the shared scale (summary page). L0 is zero everywhere.

Level L5 60 iterations, feature pins 30°

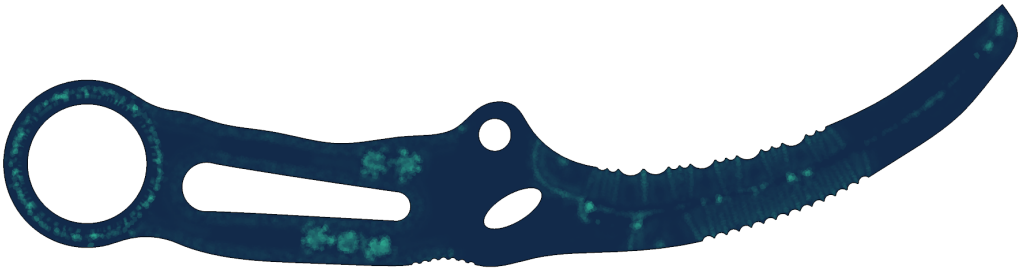
$F = 75.451120$ (with B-rep kept: 80.451120) max displacement 0.297 mm, RMS 0.0231 mm tip SF 0.6203 3D mass 83.4291 g



Whole design, oblique view, raking light, thickness direction exaggerated 2.5×; colour is height above the mid-plane (scale on the summary page).



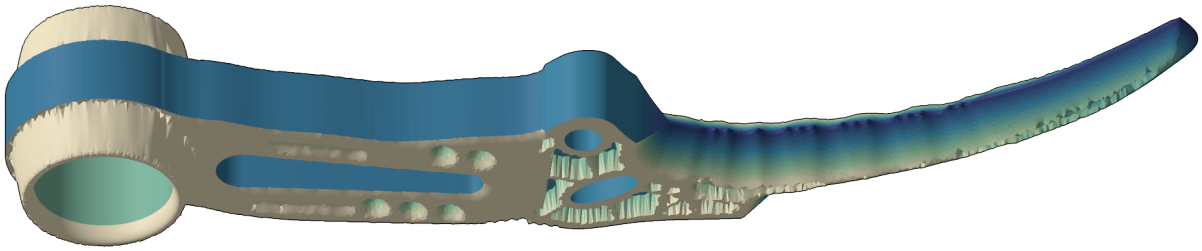
Close-ups with the same camera rotation, light and colour: guard and pivot zone (isogrid pockets, fuller windows), left; blade (grinds, serrations, tip), right.



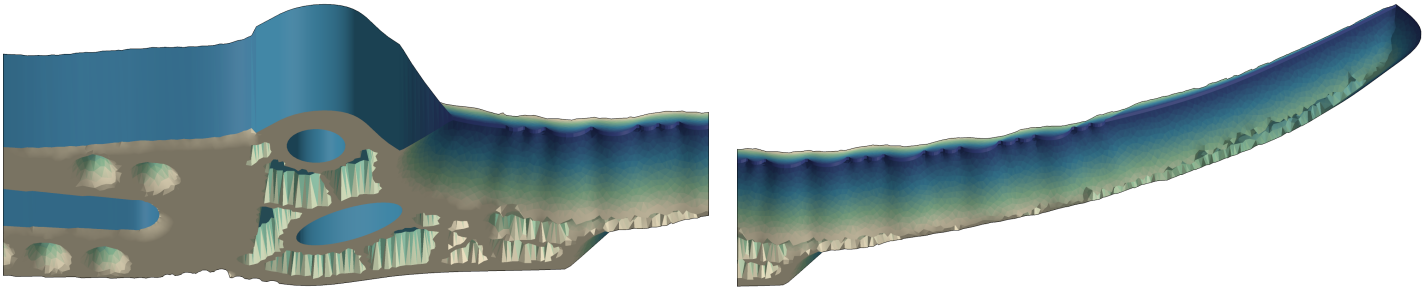
Top view; colour is displacement from L0 on the shared scale (summary page). L0 is zero everywhere.

Level L6 20 iterations, feature pins 60°

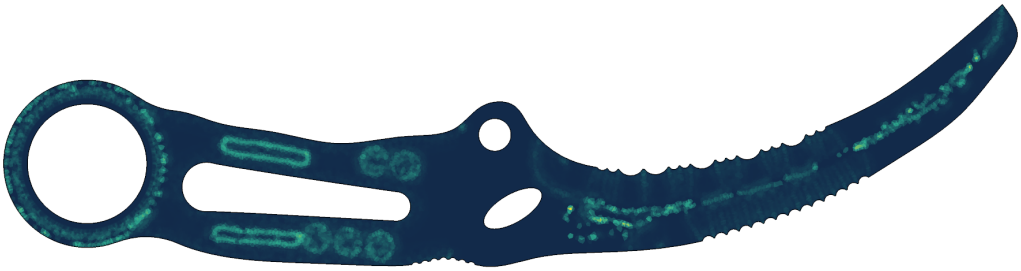
$F = 75.643828$ (with B-rep kept: 80.643828) max displacement 0.503 mm, RMS 0.0366 mm tip SF 0.6957 3D mass 83.5585 g



Whole design, oblique view, raking light, thickness direction exaggerated 2.5×; colour is height above the mid-plane (scale on the summary page).



Close-ups with the same camera rotation, light and colour: guard and pivot zone (isogrid pockets, fuller windows), left; blade (grinds, serrations, tip), right.



Top view; colour is displacement from L0 on the shared scale (summary page). L0 is zero everywhere.